STAT452
Logistic Regression and Classification

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Outline

Classification: Definition

- Given a set of records (called the *training set*)
- Each record contains a set of *attributes*. One of the attributes is the *class*.
- Find a model for the *class attribute* as a function of the values of other attributes.
- **Goal**: Previously unseen records should be assigned to a class as accurately as possible.
- Usually, the given data set is divided into training and test set, with training set used to build the model and test set used to validate it. The accuracy of the model is determined on the test set.

Classification: Definition

Consider a *qualitative target or response variable* Y and associated predictor variables $X_1, X_2, \ldots, X_p$.

The classification task is to **build a function or a rule set in terms of** $X_1, X_2, \ldots, X_p$ that takes as input and predicts its value (or category) for Y.

Examples: Qualitative variables take values in an unordered set C, such as:

- Eye color $\in$ {brown, blue, green}
- Species $\in$ {versicolour, verginica, sethosa}
- Insurance claim $\in$ {fraudulent, legitimate}

Note that these Qualitative variables take values in an unordered set C*.*

Classification Problems

Classification problems occur often, perhaps even more so than regression problems.

Some examples include:

1. A person arrives at the emergency room with a set of symptoms that could possibly be attributed to one of three medical conditions.
   Which of the three conditions does the individual have?
2. An online banking service must determine whether a transaction is fraudulent based on the user's IP address, transaction history, etc.
3. A biologist analyzing DNA sequences wants to determine which mutations are disease-causing and which are not.

Example: Exploring the Default Dataset in R

- Load dataset from the ISLR package:

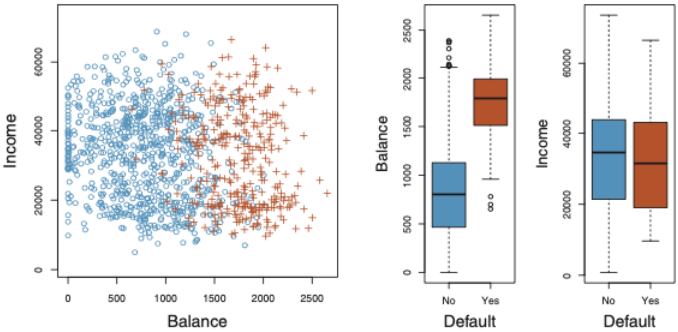
```
library('ISLR')
attach(Default)
dim(Default)
# [1] 10000     4
```

- Dataset has 10,000 observations and 4 variables.
- Preview of the dataset:

```
head(Default)
#  default student  balance    income
#1      No      No   729.53  44361.62
#2      No     Yes   817.18  12106.13
#3      No      No  1073.55  31767.14
#4      No      No   529.25  35704.49
#5      No      No   785.66  38463.50
#6      No     Yes   919.59   7491.56
```

Example: Exploring the `Default` Dataset in R

- Goal: Predict whether an individual will **default** on a credit card payment.
- Predictors: **Annual income** and **monthly credit card balance**.
- Data: Simulated dataset of 10,000 individuals.
  - Left panel: Scatter plot — defaulted (orange) vs not (blue).
  - Right panel: Boxplots showing distribution of `balance` and `income` by default status.
- Observation: Defaulters tend to have **higher balances**.
- Since `default` (Y) is qualitative, linear regression is not appropriate.



Can we use Linear Regression when Y is qualitative?

- To build a linear regression model between `default` and `balance`, we need to encode the default response into numbers:
- **Set Code:**
  - $Y = 0$, if default is `No`
  - $Y = 1$, if default is `Yes`
- **Question:** Can we simply perform a linear regression of Y on X and classify as 'Yes' if $\hat{Y}_i > 0.5$?

Limitations of using Linear Regression when Y is binary

Consider Y is `default` and X is `balance`.

In simple linear regression model: $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ or

$$E(Y_i) = \beta_0 + \beta_1 x_i.$$

For binary outcomes:

$$E(Y_i) = P(Y = 1) \cdot 1 + P(Y = 0) \cdot 0 = P(Y = 1) = P(\text{Default}).$$

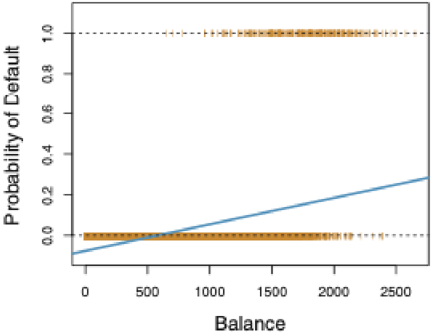
This implies

$$P(\text{Default}) = \beta_0 + \beta_1 x_i = E(Y_i).$$

NOTE: Linear regression might produce probabilities less than 0 or greater than 1.

Limitations of using Linear Regression when Y is binary

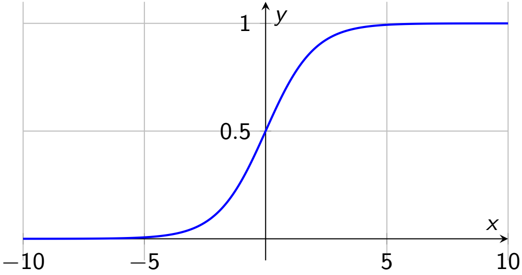
- Classification using the Default data: estimated probability of default using linear regression. Some estimated probabilities are negative!



What is Logistic Regression?

- A statistical model for predicting a binary outcome.
- Appropriate when the response variable $Y \in \{0, 1\}$.
- Models the probability $\pi(x) = P(Y = 1 \mid X = x)$ directly.
- The Logistic Model use logistic function. It maps any real-valued number into the interval $(0, 1)$.

$$y = \frac{e^x}{1 + e^x}.$$



The Logistic Regression Model

- The model form:

$$\log\left(\frac{\pi(\mathbf{X})}{1-\pi(\mathbf{X})}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = \mathbf{X}^T\boldsymbol{\beta}$$

- Equivalently, in probability scale:

$$\pi(\mathbf{X}) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}} = \frac{e^{\mathbf{X}^T\boldsymbol{\beta}}}{1+e^{\mathbf{X}^T\boldsymbol{\beta}}}.$$

Estimating the Model

For a sample of size n, the likelihood for a binary logistic regression is given by:

$$L(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} = \prod_{i=1}^{n} \left(\frac{\exp(\mathbf{X}_i^T \boldsymbol{\beta})}{1+\exp(\mathbf{X}_i^T \boldsymbol{\beta})} \right)^{y_i} \left(\frac{1}{1+\exp(\mathbf{X}_i^T \boldsymbol{\beta})} \right)^{1-y_i}.$$

This yields the log-likelihood:

$$\begin{aligned} \ell(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \left[y_i \log(\pi_i) + (1-y_i)\log(1-\pi_i) \right] \\ &= \sum_{i=1}^{n} \left[y_i \mathbf{X}_i^T \boldsymbol{\beta} - \log(1+\exp(\mathbf{X}_i^T \boldsymbol{\beta})) \right]. \end{aligned}$$

Maximizing the likelihood (or log-likelihood) has no closed-form solution, so a technique like *iteratively reweighted least squares (IRLS)* is used to find an estimate of the regression coefficients, $\hat{\boldsymbol{\beta}}$.

Odds and Log Odds

There are algebraically equivalent ways to write the logistic regression model.

First, express odds:

$$\frac{\pi}{1-\pi} = \exp(\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p).$$

- The odds is the ratio of the probability of success to the probability of failure.
- Example: If the probability of success is 0.8, then odds $= 0.8/(1-0.8) = 4$, i.e., 4:1.

Second, take the log of odds:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p.$$

- This is known as the **logit transformation**.
- The log of the odds is a linear function of the predictors.

Odds: Example and Interpretation

Definition: Odds $= \dfrac{P(\text{success})}{P(\text{failure})} = \dfrac{\pi}{1-\pi}$.

Example: In a study, 100 individuals are surveyed to check if they default on credit:

- 20 individuals defaulted $\Rightarrow P(\text{default}) = 0.2$.
- So, $P(\text{no default}) = 0.8$.
- **Odds of default** $= \dfrac{0.2}{0.8} = 0.25$.

Interpretation:

- Odds $= 0.25$ means: for every 1 person who defaults, there are 4 who do not.
- If the probability of success is 0.5, then Odds $= 1$ (equal chance).
- Odds > 1: Success is more likely than failure.
- Odds < 1: Failure is more likely than success.

Interpreting the Odds Ratio

- We compare two sets of predictors $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$, which differ in only one predictor (i.e., one predictor changes by one unit).
- This lets us assess how that single predictor affects the response.
- The **odds ratio** can take any nonnegative value:
  - If $\text{OR} = 1$: no association between predictor and response.
  - If $\text{OR} > 1$: higher predictor values increase the odds of success.
  - If $\text{OR} < 1$: higher predictor values decrease the odds of success.
- Specifically, the odds increase **multiplicatively** by $\exp(\beta_j)$ for every one-unit increase in X_j.
- This applies to both continuous predictors and categorical levels of factors.
- Values further from 1 indicate a **stronger degree of association**.

Interpreting Log-Odds in Logistic Regression

- Logistic regression models the **log-odds** of success as a linear combination of predictors:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

- Each coefficient β_j represents the change in log-odds for a one-unit increase in X_j, holding other predictors constant.
- A **positive** β_j increases the log-odds $\Rightarrow$ higher probability of success.
- A **negative** β_j decreases the log-odds $\Rightarrow$ lower probability of success.
- Exponentiating β_j gives the **odds ratio**: $\exp(\beta_j)$

Computing Odds Ratio in R: "defaut" dataset

Given a logistic regression model in R:

```
model <- glm(default ~ balance + income,
data = Default, family = "binomial")
summary(model)
```

To get the **odds ratios**:

```
exp(coef(model))
```

To get **confidence intervals** for the odds ratios:

```
exp(confint(model))
```

Each odds ratio shows the multiplicative change in odds for a one-unit increase in the predictor.

Interpreting Logistic Regression Output

We fit a logistic regression model using the `Default` dataset to predict the probability of `default = Yes` based on `balance`, i.e. an increase in `balance` is associated with an increase in the probability of `Default` .

Estimated model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \cdot \texttt{balance}.$$

Estimated coefficients:

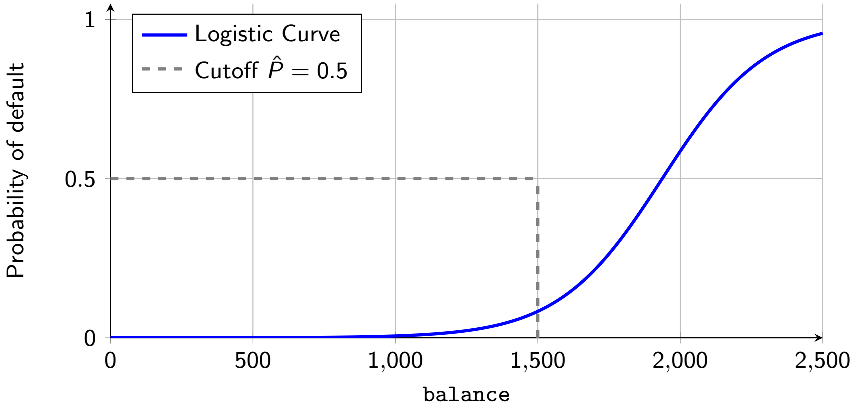
Variable	**Coefficient**	**Std. error**	**Z-statistic**	**P-value**
`Intercept`	−10.6513	0.3612	−29.5	<0.0001
`balance`	0.0055	0.0002	24.9	<0.0001

Table 1: Estimated coefficients of the logistic regression model that predicts the probability of default using balance.

Interpreting the Odds Ratio

- The estimated coefficient for `balance` is
$$\hat{\beta}_1 = 0.0055.$$
- We interpret this in terms of the **odds ratio**:
$$\text{Odds Ratio} = \exp(\hat{\beta}_1) = \exp(0.0055) \approx 1.0055.$$
- **Interpretation:**
  - For each additional unit increase in `balance`, the odds of default increase by approximately **0.55%**.
  - That is, the odds are multiplied by 1.0055 for every 1-unit increase in `balance`.
- Since the odds ratio is greater than 1, `balance` is positively associated with the likelihood of default.

Effect of Balance on Probability of Default



Statistical Significance of Coefficients: Wald Test

The Wald test is used to test the significance of individual regression coefficients in logistic regression (similar to the *t*-test in linear regression).

For maximum likelihood estimates, the test statistic is:

$$Z = \frac{\hat{\beta}_i}{\mathrm{SE}(\hat{\beta}_i)}.$$

- Used to test the null hypothesis $H_0 : \beta_i = 0$.
- Z is compared to the standard normal distribution to compute the p-value.
- Variables with small p-values are more likely to be significant predictors.

In the "defaut" data, since the p-value for `balance` is tiny (<0.0001), we **reject** H_0. We conclude that there is a statistically significant association between `balance` and probability of `default`.

Confidence Interval:

$$\hat{\beta}_i \pm z_{1-\alpha/2} \cdot \mathrm{SE}(\hat{\beta}_i).$$

Making Predictions

Using estimated coefficients from Table 1:

$$\hat{p}(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}} = \frac{e^{-10.6513+0.0055\cdot 1000}}{1+e^{-10.6513+0.0055\cdot 1000}} \approx 0.00576.$$

- For a balance of \$1,000, the predicted probability of default is ≈ 0.00576 (less than 1%)
- For a balance of \$2,000, the predicted probability is:

$$\hat{p}(X = 2000) \approx 0.586 \quad \text{(or 58.6\%)}.$$

- Shows how default risk increases non-linearly with balance.

Prediction using Categorical Predictors

Using a logistic model with a qualitative predictor `student`:

- Student variable coded as: 1 = student, 0 = non-student.
- Estimated model:

Variable	Coefficient	Std. error	Z-statistic	P-value
`Intercept`	−3.5041	0.0707	−49.55	<0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

Table 2: Logistic regression predicting probability of `default` using `student` status as a dumm variable.

$$\log\left(\frac{\pi}{1-\pi}\right) = -3.5041 + 0.4049 \cdot \texttt{student}.$$

- Predicted probabilities:

$$\hat{p}(\text{default}|\text{student} = \text{Yes}) = \frac{e^{-3.5041+0.4049}}{1+e^{-3.5041+0.4049}} \approx 0.0431$$

$$\hat{p}(\text{default}|\text{student} = \text{No}) = \frac{e^{-3.5041}}{1+e^{-3.5041}} \approx 0.0292$$

- **Conclusion:** Students have higher predicted probability of default than non-students.

Evaluating Logistic Regression: Likelihood & Deviance

- **1. Log-Likelihood** $\ell(\boldsymbol{\beta})$
  - Measures goodness-of-fit
  - Higher $\ell(\boldsymbol{\beta})$ implies better model fit.
- **2. Deviance:**

$$\text{Deviance} = -2 \cdot \ell(\text{model}) + 2 \cdot \ell(\text{saturated model}).$$

  - A lower deviance indicates a better fit
  - Deviance of the null model (intercept-only) is used as a baseline
- **3. Likelihood Ratio Test (LRT):**

$$G^2 = -2 \cdot (\ell_{\text{null}} - \ell_{\text{model}}) \sim \chi^2_{df}.$$

  - Compares two nested models
  - Test if adding predictors significantly improves model fit

Evaluating Logistic Regression: AIC and Pseudo-R^2

- **4. AIC (Akaike Information Criterion):**

$$\text{AIC} = -2\ell(\boldsymbol{\beta}) + 2p.$$

    - Balances model fit and complexity
    - Penalizes overfitting
    - Lower AIC indicates a better model

- **5. Pseudo-R^2 (e.g., McFadden's):**

$$R^2 = 1 - \frac{\ell_{\text{model}}}{\ell_{\text{null}}}.$$

    - Measures improvement over null model
    - Interpretation is less direct than in linear regression
    - Commonly used to communicate model quality

Evaluating Logistic Regression in R (Default Data)

Fit model:

```
model1 <- glm(default ~ balance, data = Default, family = "binomial")
summary(model1)
```

Selected output:

- Residual deviance: 2908.7 on 9998 degrees of freedom
- Null deviance: 2920.6 on 9999 degrees of freedom
- AIC: 2912.7
- p-value for balance $< 2e - 16 \Rightarrow$ highly significant

Likelihood Ratio Test:

```
model0 <- glm(default ~ 1, data = Default, family = "binomial")
anova(model0, model1, test = "Chisq")
```

- Compare null vs. full model
- Deviance reduction: 1324.1 ($p < 2e - 16$)
- Conclusion: balance improves model fit significantly

Comparing Logistic Models in R

We compare two nested models:

```
model1 <- glm(default ~ balance, data = Default, family = "binomial")
model2 <- glm(default ~ balance + income, data = Default,
                                                     family = "binomial")
```

Step 1: Compare AIC values

- model1 AIC $= 1600.5$
- model2 AIC $= 1585 \Rightarrow$ model2 is slightly better (lower AIC)

Step 2: Likelihood Ratio Test (LRT)

```
 >  anova(model1, model2, test = "Chisq")
  Analysis of Deviance Table
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1       9998      1596.5
2       9997      1579.0  1   17.485 2.895e-05 ***
```

- Test statistic $= 17.485$, df $= 1$, $p = 2.895 \times 10^{-5}$
  $\Rightarrow$ significant improvement by adding income.

Model Selection: AIC vs. Likelihood Ratio Test

Two common criteria:

- **AIC (Akaike Information Criterion):**
  - Balances goodness-of-fit and model complexity
  - Lower AIC = better model (penalizes extra predictors)
  - Can compare non-nested models
- **Likelihood Ratio Test (LRT):**
  - Formal hypothesis test for nested models
  - Tests if additional predictors improve the model significantly
  - Based on χ^2 distribution

When they disagree:

- **AIC prefers** more complex model, but **LRT is not significant:**
  - Extra predictor does not improve fit enough to justify complexity
  - Prefer simpler model (especially for interpretability)

Guiding principle: Use LRT for significance, AIC for prediction.

Classification Using Logistic Regression

- Logistic regression predicts the **probability** that $Y = 1$, given X:

$$\pi(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

- To classify a new observation:
  - If $\pi(X) \geq 0.5 \rightarrow$ predict class 1 (success)
  - If $\pi(X) < 0.5 \rightarrow$ predict class 0 (failure)
- This threshold of 0.5 can be changed depending on:
  - Application context (e.g., fraud detection, medical diagnosis)
  - Desired trade-off between **false positives** and **false negatives**
- The boundary $\pi(X) = 0.5$ corresponds to:

$$\log\left(\frac{\pi}{1-\pi}\right) = 0 \quad \text{implies} \quad \beta_0 + \beta_1 X = 0$$

$\rightarrow$ A linear decision boundary in predictor space.

Confusion Matrix

Definition: A table showing predicted vs. actual classifications:

	Predicted: 1	**Predicted: 0**
Actual: 1	True Positive (TP)	False Negative (FN)
Actual: 0	False Positive (FP)	True Negative (TN)

Goal: Maximize TP and TN, minimize FP and FN

- Change the threshold, 0.5 by defaut, to maximize TP and TN.
- Different applications value different types of errors (e.g. fraud, medical)

Classification Metrics

Based on Confusion Matrix:

- **Accuracy:**

$$\frac{TP + TN}{TP + TN + FP + FN}.$$

- **True Positive Rate (Sensitivity or Recall):**

$$\text{TPR} = \frac{TP}{TP + FN}.$$

- **False Positive Rate:**

$$\text{FPR} = \frac{FP}{FP + TN}.$$

- **Precision:**

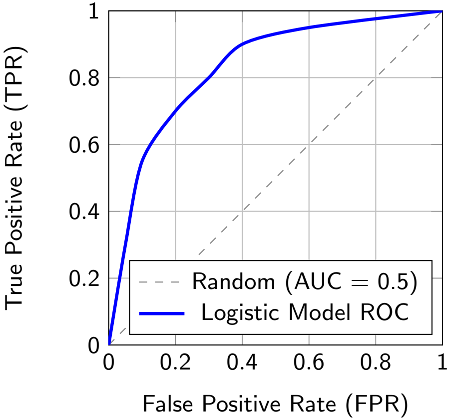
$$\text{Precision} = \frac{TP}{TP + FP}.$$

- **F1 Score:** Harmonic mean of Precision and Recall

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

ROC Curve (Receiver Operating Characteristic)

- ROC curve plots TPR (sensitivity) vs. FPR (1 - specificity) across thresholds. ROC helps visualize the trade-off between Sensitivity and Specificity.
- The closer the curve to the top-left corner, the better the model. A perfect classifier hugs the top-left corner.
- A random classifier lies along the diagonal



AUC and Threshold Selection

- **AUC = Area Under the ROC Curve**
    - Measures the ability of the model to discriminate between classes
    - AUC = 1: perfect model, AUC = 0.5: no discrimination (random).
- **Choosing Optimal Threshold τ:**
    - Depends on the cost of false positives and false negatives
    - Often chosen to maximize:

$$\text{Youden's Index} = \text{TPR} - \text{FPR}.$$

    - Or to balance Precision and Recall (F1 score)
- Tools in R: `ROCR`, `pROC`, `yardstick` (tidymodels).

Classification with Logistic Regression (Default Dataset)

Step 1: Fit model and predict probabilities

```
library(ISLR)
model <- glm(default ~ balance, data = Default, family = "binomial")
probs <- predict(model, type = "response")
```

Step 2: Classify with threshold $\tau = 0.5$

```
pred <- ifelse(probs > 0.5, "Yes", "No")
```

Step 3: Confusion Matrix and Evaluation

```
table(Predicted = pred, Actual = Default$default)
```

Step 4: Compute metrics

```
library(caret)
confusionMatrix(as.factor(pred), Default$default, positive = "Yes")
```

Classification Evaluation Example (R Output)

- Example Confusion Matrix:

	Actual: No	**Actual: Yes**
Pred: No	9625	233
Pred: Yes	42	100

- **Accuracy** $\approx 97.25\%$
- **Sensitivity (TPR)** $\approx 3.0\ \%$
- **Specificity (TNR)** $\approx 99.57\%$
- **Precision** $\approx 70.42\%$
- **Conclusion:** Good overall accuracy, but very poor sensitivity $\rightarrow$ default cases are rare and hard to detect with 0.5 threshold.

Adjusting the Threshold τ

- Default threshold $\tau = 0.5$ gave poor sensitivity
- Try lower threshold to capture more defaulters:

```
pred_30 <- ifelse(probs > 0.3, "Yes", "No")
table(Predicted = pred_30, Actual = Default$default)
confusionMatrix(as.factor(pred_30), Default$default, positive = "Yes")
```

- Lowering τ increases True Positives (TP)
- But also increases False Positives (FP)
- Trade-off: better sensitivity vs. worse specificity

ROC Curve and AUC for Model Evaluation

Using the `pROC` package:

```
library(pROC)
roc_obj <- roc(Default$default, probs)
plot(roc_obj, col = "blue", lwd = 2)
auc(roc_obj)
```

- **ROC curve:** shows performance across all thresholds
- **AUC:** Area Under the Curve
  - AUC $= 0.5$: random guessing
  - AUC $= 1.0$: perfect model
  - AUC > 0.8: strong discrimination
- Use ROC to choose threshold that balances TPR and FPR.

Choosing Threshold from ROC Curve

- Each point on the ROC curve corresponds to a threshold τ
- Changing τ changes:
  - TPR (sensitivity)
  - FPR (1 - specificity)
- **Common strategy:** Choose τ to maximize **Youden's Index**:

$$\text{Youden} = \text{TPR} - \text{FPR}$$

- This maximizes the vertical distance from the diagonal (random guessing)
- Alternative: choose τ that balances Precision and Recall (maximize F1)
- R function with `pROC`:

```
coords(roc_obj, "best", best.method = "youden")
```

Optimal Threshold via ROC (Youden)

Step-by-step in R:

```
library(ISLR)
library(pROC)

# Fit logistic regression model
model <- glm(default ~ balance, data = Default, family = "binomial")
probs <- predict(model, type = "response")

# Compute ROC object
roc_obj <- roc(Default$default, probs)

# Plot ROC curve
plot(roc_obj, col = "blue", lwd = 2)

# Find optimal threshold (Youden's Index)
coords(roc_obj, "best", best.method = "youden")
```